MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **WEEKLY TEST – 01** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 4 Multiple Choice Questions and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Short Answer Type-I question of 2 marks.
4. **Section C** comprises 3 Short Answer Type-II questions of 3 marks each.
5. **Section D** comprises 1 Applied Case-Based question of 4 marks with internal subparts.

SECTION A: Objective Type Questions (5 × 1 = 5 Marks)

Q1. If $\cos A = \frac{4}{5}$, then the value of $\tan A$ is: [1]

(A) $\frac{3}{5}$ (B) $\frac{3}{4}$ (C) $\frac{4}{3}$ (D) $\frac{5}{3}$

Q2. If $\sin\theta = \frac{a}{b}$, then $\cos\theta$ is equal to: [1]

(A) $\frac{b}{\sqrt{b^2-a^2}}$ (B) $\frac{b}{a}$
(C) $\frac{\sqrt{b^2-a^2}}{b}$ (D) $\frac{a}{\sqrt{b^2-a^2}}$

Q3. If $\tan\theta = \frac{1}{\sqrt{3}}$, then the value of $\frac{\csc^2\theta - \sec^2\theta}{\csc^2\theta + \sec^2\theta}$ is: [1]

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{3}{4}$ (D) $\frac{2}{3}$

Q4. If $3\cot\theta = 2$, then the value of $\frac{4\sin\theta - 3\cos\theta}{2\sin\theta + 6\cos\theta}$ is: [1]

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{2}$ (D) $\frac{3}{4}$

Q5. Assertion (A): The value of $\sin\theta = \frac{4}{3}$ is not possible for any acute angle θ.
Reason (R): Hypotenuse is the longest side in a right-angled triangle, so $\sin\theta = \frac{\text{Perpendicular}}{\text{Hypotenuse}} \leq 1$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Short Answer Type-I (1 × 2 = 2 Marks)

Q6. If $\sin\theta = \frac{12}{13}$, evaluate the value of $\frac{2\cos\theta + 3\tan\theta}{\sin\theta}$. [2]

SECTION C: Short Answer Type-II (3 × 3 = 9 Marks)

Q7. Prove that: $(\csc\theta - \cot\theta)^2 = \frac{1-\cos\theta}{1+\cos\theta}$. [3]

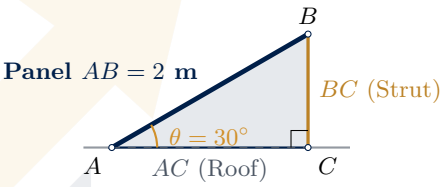
Q8. Prove that: $\frac{\cos A}{1+\sin A} + \frac{1+\sin A}{\cos A} = 2\sec A$. [3]

Q9. In ΔPQR, right-angled at Q, $PR+QR = 25$ cm and $PQ = 5$ cm. Determine the values of $\sin P, \cos P$, and $\tan P$. [3]

SECTION D: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q10. Rooftop Solar Photovoltaic Panel Array Tilt Angle

A solar engineering firm installs rooftop photovoltaic panels. For maximum solar energy absorption at a specific latitude, the panels must be mounted at an angle of $\theta = 30°$ to the horizontal roof. The slant length of each panel module is $AB = 2.4$ m. A vertical strut BC supports the upper edge, while the panel rests on the roof at A, forming a right-angled triangle ABC with $\angle C = 90°$.



Based on the above geometric model, answer the following questions:

(i) Write the trigonometric ratio connecting vertical height BC, slant length AB, and tilt angle θ. [1]

(ii) Calculate the exact height BC of the vertical support strut. [1]

(iii) Find the horizontal roof space AC covered by the panel, and determine the ratio $\frac{BC}{AC}$. [2]

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